

Consider a hyperbolic encounter involving planets 1 and 2. Assume that the motion is in the x - y plane, and the angular momentum of the relative motion points in the positive z direction. The initial obliquity of planet 1 is given by the unit vector $\boldsymbol{\varepsilon} = \varepsilon_x \hat{\mathbf{x}} + \varepsilon_y \hat{\mathbf{y}} + \varepsilon_z \hat{\mathbf{z}}$. For an encounter with eccentricity $e > 1$, the change in obliquity of planet 1 due the torque from planet 2 is approximately

$$\Delta \boldsymbol{\varepsilon} = 3 \left(\frac{J_2 + q}{\lambda + l} \right)_1 \frac{GM_2}{\omega_1 s^3} \boldsymbol{\tau} , \quad (1)$$

where $\boldsymbol{\tau}$ is a vector given by

$$\boldsymbol{\tau} = \frac{s}{u_s} \frac{1}{(1+e)} \left\{ (-\varepsilon_x \varepsilon_z \hat{\mathbf{y}} + \varepsilon_x \varepsilon_y \hat{\mathbf{z}}) \left[\cos^{-1} \left(-\frac{1}{e} \right) - \frac{1}{3} e \left(1 - \frac{1}{e^2} \right)^{1/2} \left(\frac{1}{e^2} - 4 \right) \right] + \right. \\ \left. (\varepsilon_y \varepsilon_z \hat{\mathbf{x}} - \varepsilon_y \varepsilon_x \hat{\mathbf{z}}) \left[\cos^{-1} \left(-\frac{1}{e} \right) + \frac{1}{3} e \left(1 - \frac{1}{e^2} \right)^{1/2} \left(\frac{1}{e^2} + 2 \right) \right] \right\} . \quad (2)$$

The magnitude of $\boldsymbol{\tau}$ is an effective timescale over which the torque acts. Here s and u_s are, respectively, the separation and relative speed at closest approach. The eccentricity is related to the Safronov parameter, $\theta = 2G(M_1 + M_2)/(su^2)$, by

$$e = 1 + 2\theta^{-1} . \quad (3)$$

When $\varepsilon_y = 0$ and $\varepsilon_x = \pm \varepsilon_z = \pm 1/\sqrt{2}$, we obtain the maximum obliquity change for a given e , where the corresponding $\boldsymbol{\tau}$ is

$$\boldsymbol{\tau}_{\max} = \pm \frac{1}{2} \frac{s}{u_s} \frac{1}{(1+e)} \left[\cos^{-1} \left(-\frac{1}{e} \right) - \frac{1}{3} e \left(1 - \frac{1}{e^2} \right)^{1/2} \left(\frac{1}{e^2} - 4 \right) \right] \hat{\mathbf{y}} . \quad (4)$$

In the limit of a straight-line trajectory ($e \rightarrow \infty$), s and u_s become the impact parameter b and speed at infinity u , and

$$|\boldsymbol{\tau}_{\max}|_{e \rightarrow \infty} = \frac{2}{3} \frac{b}{u} . \quad (5)$$

For a parabolic encounter ($e \rightarrow 1$), we have $u_s^2 = 2G(M_1 + M_2)/s$, and

$$|\boldsymbol{\tau}_{\max}|_{e \rightarrow 1} = \frac{\pi}{4} \frac{s}{u_s} = \frac{\pi}{4} \left[\frac{s^3}{2G(M_1 + M_2)} \right]^{1/2} . \quad (6)$$